

LETLHAKANE, Botswana

WMO: 680400

Lat: 21.417S

Lon: 25.600E

Elev: 3232

StdP: 13.06

Time Zone: 2.00 (E02)

Period: 96-15

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	41.0	43.7	17.6	15.0	58.6	21.4	18.0	57.9	19.0	64.2	17.5	64.5	4.7	160	0.364

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11	23.7	98.4	66.0	96.7	66.7	94.9	66.7	73.4	85.0	72.5	84.3	71.8	83.7	7.1	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.4	126.4	76.4	69.8	123.8	75.7	68.8	119.5	75.5	39.3	84.9	38.5	84.1	37.8	83.7	78.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.4	16.4	14.3	DB	35.4	101.9	3.2	1.6	33.1	103.1	31.3	104.1	29.5	105.0	27.2	106.2	(4)
(5)			WB	29.4	76.3	2.7	1.2	27.5	77.2	25.9	77.9	24.4	78.6	22.4	79.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	72.7	79.4	78.8	76.5	72.3	66.0	61.9	60.7	66.5	73.1	78.6	79.6	79.2	(6)	
	DBStd	8.05	3.98	3.27	4.07	3.97	4.35	4.01	3.52	4.96	4.80	4.39	4.66	3.72	(7)	
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD65	335	0	0	1	3	39	108	140	40	4	0	0	0	(9)	
	CDD50	8278	911	807	821	669	495	358	330	512	694	888	887	906	(10)	
	CDD65	3139	446	387	357	222	70	16	5	87	248	423	437	441	(11)	
	CDH74	35821	4614	3812	3424	2368	1261	419	392	1447	3190	5193	5137	4564	(12)	
	CDH80	16747	2233	1786	1534	918	333	61	50	464	1581	2847	2722	2218	(13)	
(14) Wind	WSAvg	6.1	6.0	5.8	5.6	5.5	5.0	5.5	5.6	6.4	7.3	8.0	6.9	6.2	(14)	
(15) Precipitation	PrecAvg	15.2	3.3	3.1	2.3	0.5	0.2	0.1	0.0	0.0	0.1	0.7	2.0	2.7	(15)	
	PrecMax	32.1	9.5	10.7	6.5	1.2	1.5	1.3	0.1	0.1	0.7	1.7	4.1	4.3	(16)	
	PrecMin	5.2	0.1	0.1	0.0	0.3	0.0	0.0	0.0	0.0	0.0	0.2	0.1	1.1	(17)	
	PrecStd	7.0	2.6	2.8	2.3	0.2	0.4	0.3	0.0	0.0	0.2	0.4	1.3	0.9	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	98.9	97.8	97.0	94.7	90.1	84.5	85.6	91.4	98.2	100.2	100.8	98.1	(19)	
		MCWB	69.0	69.2	68.1	68.1	61.9	58.9	57.6	61.1	63.8	64.8	66.2	67.3	(20)	
	2%	DB	96.3	95.0	94.7	90.0	86.4	81.6	81.1	87.8	94.6	98.2	97.6	95.4	(21)	
		MCWB	68.8	68.6	68.1	64.7	60.9	59.0	56.2	60.0	62.7	64.3	66.4	68.0	(22)	
	5%	DB	93.4	92.3	91.6	88.0	84.0	78.7	78.6	85.0	91.8	95.4	95.2	93.4	(23)	
		MCWB	68.5	68.9	67.5	64.3	60.5	57.9	55.4	59.6	61.7	64.8	66.3	68.1	(24)	
	10%	DB	91.1	89.9	88.2	85.8	80.9	75.9	75.4	82.2	89.4	93.3	92.9	91.2	(25)	
		MCWB	68.7	68.8	67.5	64.1	59.5	56.7	55.1	58.4	61.6	64.5	66.4	68.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	74.5	74.4	73.5	71.7	66.3	64.4	61.8	65.3	68.8	71.0	72.4	74.2	(27)	
		MCDB	85.0	84.9	85.0	85.8	78.5	77.9	78.9	83.6	88.1	90.6	88.2	86.5	(28)	
	2%	WB	73.2	73.1	71.9	69.7	63.9	62.0	58.8	62.8	66.3	69.2	71.0	72.8	(29)	
		MCDB	83.3	84.9	83.5	83.7	80.9	75.3	76.4	83.1	87.1	88.8	86.6	85.2	(30)	
	5%	WB	72.3	72.3	71.0	68.0	62.5	60.0	57.3	61.0	64.7	67.9	70.1	71.8	(31)	
		MCDB	82.8	84.0	82.7	80.3	80.3	73.8	74.2	81.2	85.7	88.0	85.8	84.4	(32)	
	10%	WB	71.4	71.4	70.2	66.3	60.9	58.1	56.0	59.5	63.2	66.8	69.1	70.9	(33)	
		MCDB	82.2	82.8	82.0	78.7	78.1	72.2	72.8	79.3	84.1	85.8	84.4	83.6	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	21.2	21.2	21.5	25.8	30.1	28.0	29.7	30.0	29.6	26.2	23.7	21.4	(35)	
		MCDBR	25.8	26.0	27.0	29.1	33.1	32.2	35.0	33.2	33.2	30.5	27.8	25.6	(36)	
	5% WB	MCWBR	8.7	8.5	9.6	11.5	14.7	15.9	17.7	15.6	14.5	11.7	9.5	7.9	(37)	
		MCWBR	19.6	19.6	20.0	23.5	29.0	26.6	29.6	30.7	28.9	25.5	22.7	21.5	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.372	0.373	0.360	0.332	0.297	0.289	0.290	0.355	0.426	0.429	0.386	0.376	(40)	
		taud	2.417	2.409	2.437	2.460	2.505	2.494	2.481	2.277	2.098	2.149	2.318	2.394	(41)	
	Ebn at Noon		308	304	300	295	295	291	295	283	275	284	302	307	(42)	
		Edh at Noon	40	40	37	33	29	28	30	39	51	51	44	41	(43)	
(44) All-Sky Solar Radiation	RadAvg		2233	2120	1982	1803	1633	1448	1541	1793	2070	2227	2228	2177	(44)	
	RadStd		153	194	149	122	46	64	41	80	94	112	148	141	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page